PHASE-2 REPORT

Data Analytics Track | Tasks 6–10

Name	Vaishnavi Raut
Track	Data Analytics – PulseIQ
GitHub Repository	https://github.com/vaishnaviraut629-de/pulseiq
Git Tag	phase-2
Commit	e5a337a
Status	Phase 2 Complete

1. Task Completion Summary

Task	Status	Outcome / Evidence
Task 6: Attendance	Completed	Attendance overview dashboard. Evidence: Task6_01_Overview.png
Task 7: Attrition	Completed	Attrition analytics; risk is labelled as predicted. Evidence: Task7_Attrition.png
Task 8: Team Productivity	Completed	Team dashboard covering 38 sprints. Evidence: Task8_Dashboard.png
Task 9: Payroll Reconciliation	Completed	4 bad rows removed during reconciliation. Evidence: Task9_reconciliation.png
Task 10: Budget	Completed	198 rows; 94% marked Green. Evidence: Task10_Budget_198rows.png

2. Validation and Proof of Functionality

Payroll reconciliation was checked from raw data through cleaning and database aggregation. The known bad payroll row PI-X0001 had an amount of 9,999,999 in the raw data. The find_bad.py check removed 4 bad rows.

The cleaned payroll total was calculated using SUM in DuckDB. The result was compared with the answer key; the difference was 0. The chart is available in app.py on Page 9.

Manual verification: Engineering payroll for 2024-01 = 50,000 + 60,000 + 55,000 = 165,000. The answer key also showed 165,000, so the check passed.

3. Mini Challenge: Payroll Total Reconciliation

Check	Result
Raw data issue	PI-X0001 contained the bad amount 9,999,999.
Cleaning script	find_bad.py removed 4 bad rows.
Database calculation	SUM calculated in DuckDB.
Answer-key comparison	Difference = 0.
Manual check	Engineering, 2024-01: 50k + 60k + 55k = 165k; answer key = 165k (OK).
Dashboard evidence	app.py – Page 9.

4. Key Business Definitions

Term	Simple explanation used in the project
Reconciliation	Comparing wallet and bank counts/records to identify mismatches; 4 fake notes/rows were removed.
Variance	Difference between budget and actual spending. Example: budget 100 and actual 120 gives +20 overspend.
Green status	Green means OK under the project's budget rule; Task 10 reports 94% Green.
Predicted risk	A forecast, not a confirmed outcome. Attrition risk must remain clearly labelled “Predicted”.

5. Key Learnings

• Data cleaning is a major part of analytics work; the note from this phase is that clean data is about 70% of the work.

• Small groups should be hidden to protect privacy.

• Velocity alone can be misleading and should be interpreted with other productivity measures.

• Reconciliation and manual checks help verify that dashboard figures match the underlying data.

• Predictions should be clearly distinguished from actual results.

6. Problems Encountered and Resolutions

Issue	Resolution
Bad payroll rows	Used find_bad.py; 4 bad rows were removed.
Task 9 dashboard/page not visible	Worked through the issue with 19 commits.
Budget KPI rule	Applied the 94% Green rule for Task 10.

7. Next Phase Plan

Phase 3 will apply the same variance logic to revenue, work on DSQ, and keep all risk outputs clearly labelled as “Predicted” where applicable.

8. Screenshots and Supporting Evidence

Task	Evidence filename / note
Task 6 – Attendance	Task6_01_Overview.png
Task 7 – Attrition	Task7_Attrition.png (Predicted)
Task 8 – Team Productivity	Task8_Dashboard.png (38 sprints)
Task 9 – Payroll	Task9_reconciliation.png (4 bad rows out)
Task 10 – Budget	Task10_Budget_198rows.png (94% Green)

Repository and Submission Details

GitHub: https://github.com/vaishnaviraut629-de/pulseiq

Git tag: phase-2 | Commit: e5a337a | Status: Phase 2 Complete

Submission note: git tag phase-2, git push --tags, and upload this PDF.

End of Phase-2 Report